Gluings of fundamental groups, via theory of descent

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Abstract

One of the main guiding principles in local-to-global arguments is the following: injective maps on local data should induce an injective map on the global data. In general, this can only be guaranteed when the local injective maps agree on intersections.

We can frame the principle as follows: given a morphism $f\colon B_1 +_A B_2 \to Y_1 +_X Y_2$ in a category $\mathcal{C}$ between two pushouts (the global data) induced by a suitable pair of monomorphisms $f_i\colon B_i \to Y_i$ for $i = 1, 2$ (the local data), what conditions guarantee that f is a monomorphism as well? As a guideline, we verify that the Van Kampen condition in an adhesive category $\mathcal{C}$ [3] will ensure the principle holds.

This problem was inspired by the study of incompressible and essential surfaces in 3-manifolds [4]. For instance, the Loop Theorem [2] provides a relationship between incompressibility and injectivity of the relevant morphisms on fundamental groups; so we focus our attention on groups homomorphisms and graphs of groups.

In our application, this argument is done via two steps: (1) a strong amalgamation property [6], to guarantee that the pushout squares are pullbacks, and (2) an argument which studies the comparison between the global objects. More specifically, the category Grp of groups satisfies the strong amalgamation property, while the comparison step is supplied by a "no-fold condition" on Bass-Serre trees [1, 5].

This abstract perspective brings forth three contributions: (1) it may be applied more generally than standard arguments, (2) it arguably better captures the geometry of the gluings involved, by specifying the required no-fold conditions, and (3) it provides a tractable set of conditions, intended for future applications.

This is joint work with F. Lucatelli Nunes and J. Nogueira.

Keywords: strong amalgamation, Van Kampen pushout, amalgamated free product, graph of groups, Bass–Serre tree, π_1-injective surface.

Subject area: Category theory, geometric topology.

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